Dependency Flow Planning (DFP): A Novel Bottom-Up Methodology for Sequential Project Management

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***Abstract*—The modern landscape of project management is characterized by complex, interdependent tasks that often challenge traditional planning methodologies. While existing frameworks like the Work Breakdown Structure (WBS) and Kanban excel at task decomposition and workflow visualization, they frequently fall short in explicitly representing and enforcing the sequential dependencies inherent in many projects. This paper introduces Dependency Flow Planning (DFP), a novel methodology designed to address this gap. DFP fundamentally reverses the traditional top-down hierarchy, defining a project not by a single, large goal but by a bottom-up flow where parent tasks are the logical next steps unlocked by the completion of their child tasks.**

The methodology is grounded in three core principles: a child-to-parent flow, strict and enforced task dependencies, and a visual mapping canvas. This approach creates a clear, action-oriented roadmap that prevents common errors like starting tasks prematurely and provides an intuitive, at-a-glance view of a project's critical path. Through a comparative analysis, this paper demonstrates how DFP can be used to complement established methodologies, offering a powerful tool for strategic planning and tactical execution. The findings suggest that DFP provides a more robust and intuitive solution for managing complex, sequential projects, paving the way for improved efficiency and clearer communication.

***Index Terms*—DFP, tasks, planning, management, canvas, map, wbs, kanban, gantt, cpm, pert, methodology**

1 Introduction

1.1 Background and Context

Project management has evolved from a simple practice of organizing tasks into a sophisticated discipline for navigating complexity, resource constraints, and dynamic environments. At its core, a project is a series of interconnected activities that must be completed to achieve a specific outcome. While traditional methodologies like waterfall [1] provide a rigid, linear structure, and agile frameworks [2] [1] offer flexibility and iterative development, a persistent challenge remains: effectively representing and managing the precise sequence and dependencies of tasks.

The limitations of existing frameworks become apparent in projects where a logical, sequential flow is critical for success. The Work Breakdown Structure (WBS), for instance, is excellent for decomposition but does not inherently manage dependencies. A task in a WBS can be broken down, but its relationship to other tasks—what must be done before it can start—is not visually or functionally enforced. Similarly, Kanban boards [3] brilliantly visualize the state of a workflow but are primarily a pull system; while they can use blocking tags to show dependencies, they don't provide an integrated, strategic view of the entire project's sequential logic from the outset. This often leads to fragmented planning, where dependencies are managed ad-hoc or through separate tools, creating a disconnect between the high-level plan and the day-to-day execution. This paper addresses this gap by proposing a new methodology that makes sequential dependency the central organizing principle of the project plan.

1.2 Literature Review

The development of Dependency Flow Planning (DFP) is informed by a critical examination of established project management methodologies and their approaches to task hierarchy and dependencies. This review focuses on frameworks that have influenced DFP's core concepts, highlighting their strengths and, more importantly, their limitations in addressing the specific problem of bottom-up, sequential planning.

1.2.1 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) [4] is a foundational project management tool that decomposes a project into smaller, more manageable components. It provides a hierarchical, deliverable-oriented view of the work, allowing teams to organize and estimate effort. While the WBS is effective for defining scope and assigning responsibility, its primary limitation is its lack of built-in dependency management. A WBS tree shows what needs to be done, but not necessarily in what order. The relationships between tasks are implied or managed separately, leading to potential misalignments between planning and execution.

1.2.2 Gantt Charts and the Critical Path Method (CPM)

Gantt charts [6] and the Critical Path Method (CPM) [7] are powerful tools for visualizing project timelines and managing dependencies. A Gantt chart displays tasks as horizontal bars on a timeline, clearly showing start and end dates and their sequential dependencies. Figure 1 shows an example of a project using Gantt chart. CPM takes this a step further by

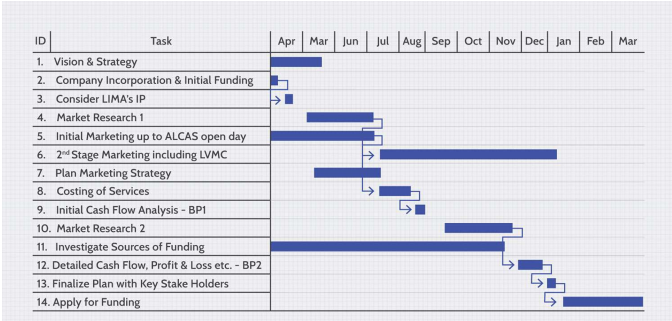

Fig. 1. Gantt Chart example. Image adapted from Investopedia [5].

identifying the critical path—the longest sequence of dependent tasks—which determines the minimum time required to complete the project. CPM's main objective is to map out all possible paths to project completion and then identify the most efficient route. As such, CPM is primarily used during the initial planning phase and is not well-suited for tracking project progress in a flexible manner, unlike methodologies such as Kanban. An example of a CPM chart is shown in Figure 2.

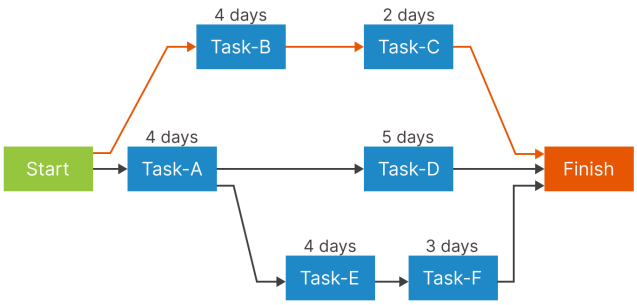

Fig. 2. Critical Path Method example.

These methodologies are highly effective for complex, linear projects with well-defined schedules. However, they can be rigid and difficult to adapt to changes. Their top-down, schedule-driven nature can also feel disconnected from the day-to-day work, as they are not inherently designed for a fluid, continuous flow.

1.2.3 Kanban and Agile Methodologies

Kanban [8], a core component of many agile frameworks, is renowned for its visual simplicity and flexibility. The Kanban board, with its columns representing workflow stages, provides an at-a-glance view of a project's status. It encourages a "pull" system, where work is initiated only when there is capacity, optimizing flow and limiting work-in-progress. While some digital Kanban tools allow for linking tasks to represent dependencies, this is often a secondary feature rather than a core organizing principle. Kanban primarily focuses on the state of a task ("To Do," "In Progress," "Done") rather

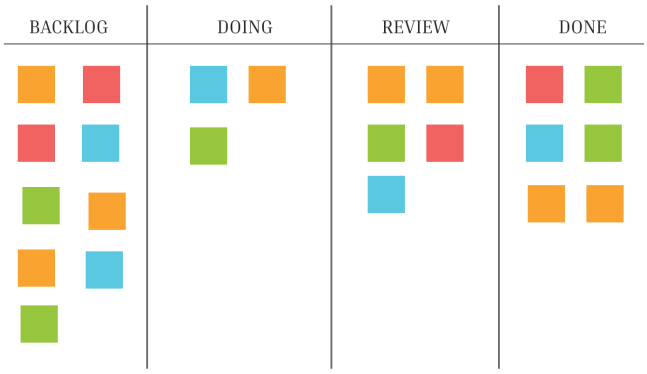

Fig. 3. Kanban example.

than its logical sequence within a broader project. This can make it challenging to gain a strategic understanding of a project's entire dependency chain at a glance. Figure 3 shows an illustrated design of how a Kanban board looks like.

1.2.4 PERT Charts

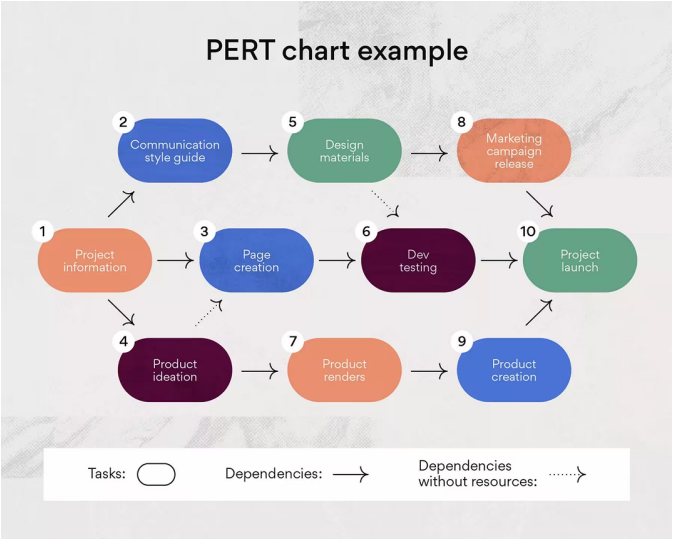

Fig. 4. PERT Chart example. Image adapted from asana [9].

Program Evaluation and Review Technique (PERT) [10] is a project management tool used for scheduling, organizing, and coordinating tasks within a project. It is particularly useful for projects with uncertain activity durations, such as research and development initiatives. A PERT chart is a visual network diagram that shows the sequence of tasks and their dependencies, much like a Gantt chart. However, its core distinction lies in its use of three-point estimation (optimistic, pessimistic, and most likely) to calculate a weighted average for task duration. This provides a more realistic and statistically-based project timeline. While PERT charts excel at managing uncertainty and providing a probabilistic view of project completion, they are complex to create and maintain. They can also be difficult

to interpret for stakeholders who are not familiar with the methodology.

PERT, by its nature, is a pre-project planning tool that requires significant upfront time for its complex calculations and network construction. While its principles can be applied to monitor and control project execution, a standardized methodology for real-time maintenance, similar to Kanban, is not universally defined. Consequently, organizations often develop their own internal variations for updating the network diagram during the project lifecycle [11]. The network diagram shown in figure 4 is another example of projects using PERT methodology for planning as well as tracking projects with its own variant by asana [9].

1.2.5 The Research Gap

The literature reveals a clear gap: no single methodology comprehensively integrates the bottom-up planning of an organic task flow with strict dependency enforcement within a single visual framework. While WBS provides hierarchy, it lacks flow. While Gantt charts provide flow, they are rigid and top-down. While Kanban provides a visual workflow, dependencies are often an afterthought. And PERT chart being the closest methodology which shares a lot of its concept with DFP, it still difficult to use and maintain. DFP is designed to fill this void by offering a methodology that is both a strategic planning tool and a tactical execution guide, making sequential dependencies the central pillar of the project plan.

1.3 Research Objectives

This paper seeks to introduce and formalize Dependency Flow Planning (DFP), a novel project management methodology designed to address the limitations identified in existing frameworks. The primary objectives of this research are as follows:

- **To Define and Articulate the Core Principles of DFP:** To clearly establish the foundational concepts of DFP, including the unique child-to-parent flow, the function of strict dependencies, and the role of the visual canvas as the primary planning interface.
- **To Explain the Practical Implementation of DFP:** To provide a step-by-step guide on how DFP can be used in a real-world project, demonstrating its application from initial planning to ongoing execution.
- **To Analyze and Compare DFP with Existing Methodologies:** To conduct a critical comparison of DFP against established frameworks such as Kanban, WBS, and Gantt charts, highlighting its unique strengths and how it can be used to complement their respective functions.
- **To Evaluate the Potential Benefits and Use Cases of DFP:** To discuss the advantages of adopting DFP, including improved clarity, communication, and project efficiency, and to identify the types of projects and industries where this methodology would be most beneficial.

1.4 Paper Structure

This paper is structured to present a comprehensive overview of the Dependency Flow Planning (DFP) methodology. Following this introduction, Section 2, ”Methodology,” will provide a detailed explanation of DFP’s core principles and implementation process. This section will be supported by visual examples to illustrate the child-to-parent flow and the visual canvas. Section 3, ”Discussion,” will then analyze DFP’s advantages, compare it with existing methodologies such as Kanban and Gantt charts, and explore its potential applications and limitations. Finally, Section 4, ”Conclusion,” will summarize the key findings, discuss the contributions of DFP to the field of project management, and suggest areas for future research.

2 Methodology: Defining Dependency Flow Planning (DFP)

2.1 Core Principles of DFP

Dependency Flow Planning (DFP) is built upon three foundational principles that together form a cohesive and intuitive methodology. These principles challenge traditional top-down planning by making sequential dependencies the central organizing element of the project.

2.1.1 The Child-to-Parent Flow

The most distinguishing principle of DFP is the reversal of the conventional task hierarchy. In most methodologies, a project is a parent goal that is broken down into smaller child tasks. DFP, conversely, defines the parent task as the next logical stage or action that is unlocked only by the successful completion of its child task(s). This creates a ”bottom-up” flow, where the project plan grows organically from the ground up, rather than being imposed from the top down. This approach ensures that every task on the map has a direct purpose: to enable a subsequent action.

For example, in a content creation project, the tasks ”Outline Article” and ”Research Keywords” are the children of the parent task ”Write First Draft.” The flow is logical: you cannot effectively write the first draft until the outline and research are complete. This model is highly effective for teams that prioritize action and sequential progression over abstract decomposition.

The nomenclature is designed to reflect this generational flow. The task a user is currently working on is referred to as the ”child task”. This task, upon completion, serves as a dependency for its ”parent task”, which represents the next stage of work. The parent task, in turn, is a dependency for a ”grandparent task”, and so on. This creates a clear, multi-generational hierarchy that visually and logically maps out the entire project’s progression from a set of starting tasks to the final deliverable. As the team progresses through the flow, the currently working tasks will become the ”child task”.

2.1.2 Strict, Enforced Dependencies

DFP operates on a strict Finish-to-Start dependency rule. A parent task cannot be started, nor can it be marked as complete, until all of its corresponding child tasks are finished. This principle is non-negotiable within the DFP framework. The strict enforcement of this rule prevents common project management pitfalls, such as tasks being started prematurely or being marked as complete when their prerequisites are not met. By baking this dependency logic directly into the planning methodology, DFP provides a robust system for preventing errors and ensuring the integrity of the project's timeline. This also makes identifying the project's critical path—the longest chain of dependent tasks—inherently visible.

This type of system is particularly effective in projects where multiple teams are working on different components in parallel. The strict dependency enforcement ensures that no single team can move on to a shared parent task until all necessary inputs from the other teams are complete. This not only highlights the precise timing of each team's progress but also prevents one team from becoming a bottleneck without the issue being immediately visible. It ensures that all contributing streams of work are synchronized before they converge, eliminating the risk of mismatched components or wasted effort.

2.1.3 Visual Canvas and Mapping

The third core principle is the use of a visual canvas as the primary interface for the project plan. DFP moves beyond linear lists or hierarchical charts, favoring a nonlinear, spatial map. Each task is represented as a node, known as task node, and the relationships between the child and parent tasks are illustrated by connecting lines or arrows. This visual representation provides an at-a-glance overview of the entire project's flow, making it easy for anyone on the team to understand dependencies and the project's current status. The visual map serves as a living document that can be updated in real time, with nodes and lines changing color or appearance to reflect task status (e.g., green for completed, gray for not started, yellow for in progress). This visual clarity significantly improves team communication and provides a clear, intuitive way to track progress.

2.2 How DFP is Implemented

The implementation of Dependency Flow Planning (DFP) is a cyclical and intuitive process that can be managed more effectively and efficiently with digital tools compared to physical tools. For complex projects and to consume the true potential of DFP, it will be advisable to prefer the digital tool over the physical one. Unlike static planning methodologies, DFP is a living framework that evolves with the project. The implementation follows a step-by-step approach designed to build the project map organically, from the first action to the final deliverable.

To demonstrate the methodology's true potential, this paper will use a digital environment specifically designed to support its principles. The visual canvas feature, known as the Map view, is an integrated component of the Task Board [12] plugin for the markdown-based note-taking application, Obsidian [13]. This environment is an ideal choice because Obsidian provides the complete framework for creating and organizing notes and tasks during the research and planning phase of a project, while the Task Board plugin centralizes these tasks for efficient management on a single board. This architecture is specifically chosen to showcase the synergistic benefits of DFP and Kanban.

The following sub-sections will elaborate on the step-by-step, generalized process that can be followed to successfully practice the DFP methodology.

2.2.1 Creating the Task Cards

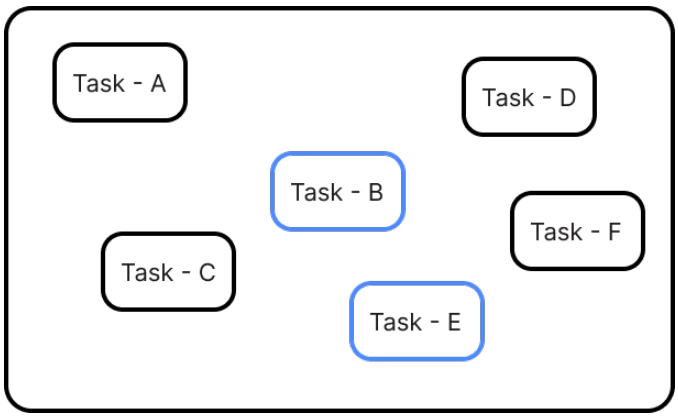

Fig. 5. Tasks collected on canvas

The initial step in implementing DFP, like many project management methodologies, involves the creation or collection of tasks, which are then represented as task cards (also known as task nodes) on the visual canvas (see Figure 5). In digital environments, the design and functionality of these cards can be highly customized to suit a team's specific needs. For example, within the Task Board plugin, the task card is designed to display a variety of crucial information (see Figure 6).

The card's components, whose visibility can be configured to minimize visual noise, include a header that provides key metadata such as task priority, tags indicating the task's domain, and the source document from which it was fetched. The main component is the task title, accompanied by a checkbox to mark completion. Below this, users can define sub-tasks to further break down the work. A detailed description field provides a space for extensive notes, links, and embedded resources. The card's footer may contain important dates, such as a due date or a scheduled time. This modular design can be adapted to any team's needs and comfort level, with buttons for navigation and editing.

This initial phase may overlap with the principles of the Work Breakdown Structure (WBS), where a project is broken down into smaller, granular, and tangible tasks. This is particularly crucial for time-constrained projects, where

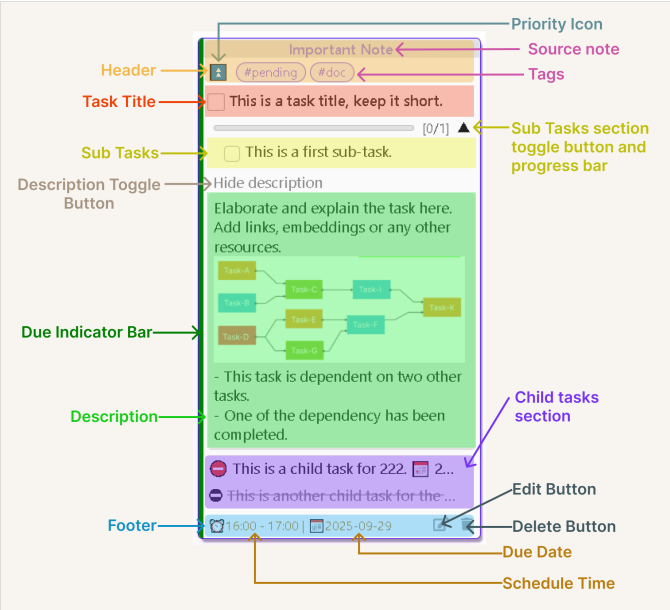

Fig. 6. Task card (also task node)

defining the final goal and outlining all steps upfront is the most efficient approach. However, for projects focused on an iterative, evolving process, the team can begin by defining only a near-term milestone and then breaking it into smaller tasks. As these milestones are completed, new tasks and their dependencies can be added accordingly.

2.2.2 Define the First Tasks

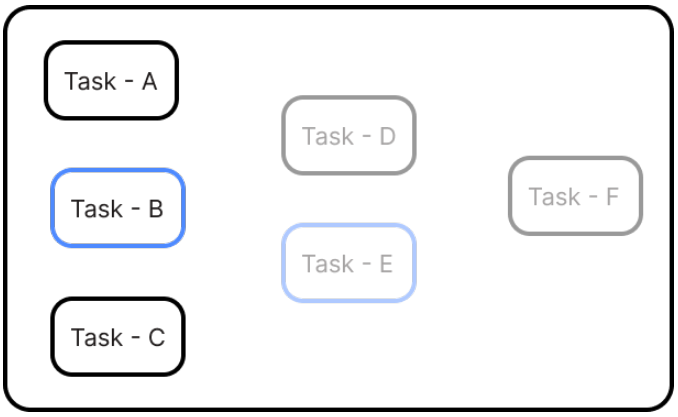

Fig. 7. Focusing on the child tasks

The DFP process begins not with a grand vision, but with the first tangible actions that need to be taken. The team identifies the initial, foundational tasks that have no preceding dependencies. These tasks represent the starting point of the project's flow and are the first nodes to be placed on the DFP canvas. For example, in a product launch, these might be "Market Research" and "Competitor Analysis."

The layout of the DFP canvas can be customized to the team's preference; for the purpose of this paper, the flow will be depicted from left to right. This chronological arrangement adds significant value by allowing the team to visually organize tasks in the order they are planned for completion. A common outcome of this step is the presence of multiple child tasks that form a column, as shown in the figure 7.

2.2.3 Map the Next Stage

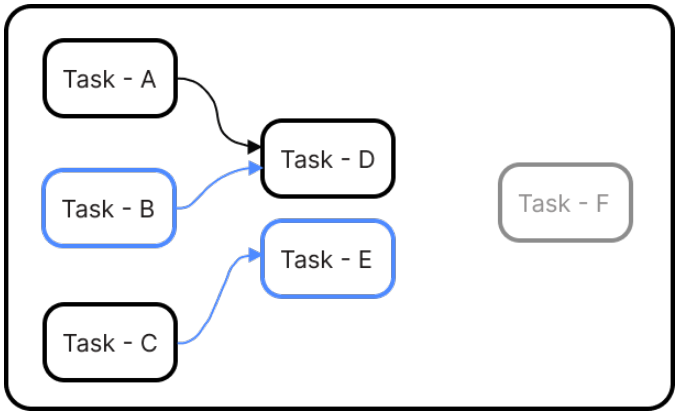

Fig. 8. Connecting child tasks to immediate parent tasks

Once the initial tasks are identified, the team collectively determines what comes next. They ask the question: "What tasks can we only start after the first tasks are complete?" The answer to this question becomes the parent task(s). For instance, after "Market Research" and "Competitor Analysis" are finished, the next logical step is "Define Product Requirements." This new task is placed on the canvas and is visually connected to its child tasks with an arrow, illustrating the flow of work (see figure 8).

2.2.4 Iterate and Build the Map

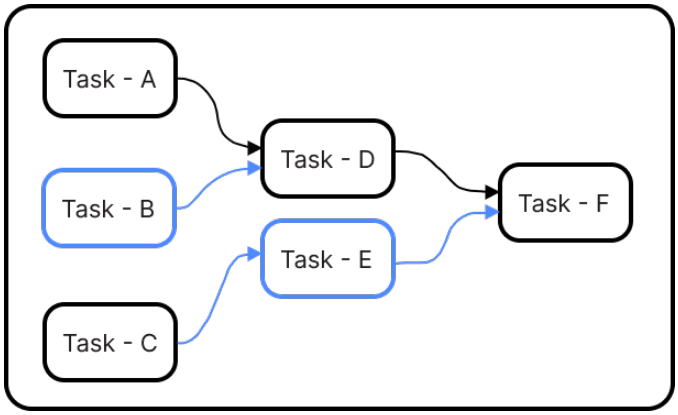

Fig. 9. Finish mapping all tasks to create a flow

This process is repeated iteratively. For each new parent task, the team identifies its own parent task—the stage that follows it—and connects them. This creates a branching and ever-expanding map that reflects the project's natural progression. New tasks can be added as new nodes at any point,

and the map can be rearranged to reflect new dependencies or changes in the project scope. The map serves as a dynamic blueprint, showing the entire project's structure as it is being built (see figure 9).

2.2.5 Execute and Progress

As the team begins work, tasks are executed and marked as complete on the DFP canvas. A key feature of DFP, especially in its digital form, is that a parent task will remain visibly "blocked" until all of its child tasks are completed. This visual cue provides clear direction for team members, indicating which tasks are ready to be worked on and which are dependent on others. The map constantly updates to reflect the project's real-time progress.

2.2.6 Identify Bottlenecks and Manage Flow

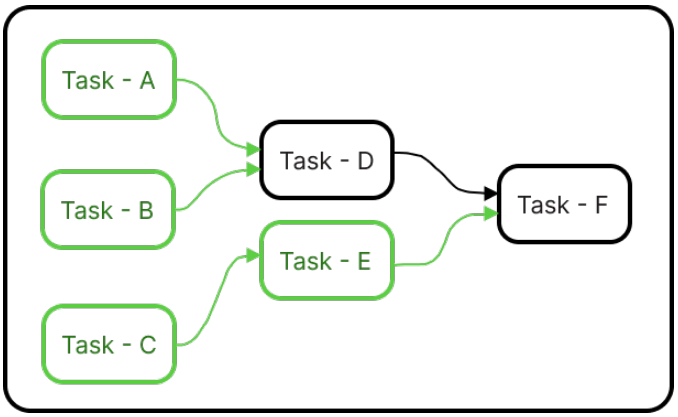

Fig. 10. Task-F has been blocked by Task-D

The visual nature of the DFP map makes it easy to identify bottlenecks. If a group of tasks is complete but their parent task hasn't been started, it's an immediate signal for action. Conversely, if a child task is stalled, the blocked parent task downstream will be clearly visible, allowing the team to quickly address the impediment (see figure 10). By focusing on the flow of work and removing blockages, the team can maintain efficiency and ensure a smooth progression toward project completion.

3 Discussion

3.1 Advantages of DFP

Dependency Flow Planning (DFP) offers a number of significant advantages over traditional methodologies, stemming from its focus on sequential logic and visual representation. These benefits translate directly to improved project outcomes, clearer communication, and more efficient team collaboration.

- **Clarity and Simplicity:** DFP eliminates the complexity of multi-layered hierarchies and abstract charts. By focusing on a straightforward child-to-parent flow, it provides a simple and intuitive roadmap that anyone on the team can immediately understand. The visual canvas makes it clear what needs to be done, in what order, and what tasks are blocked, reducing confusion and guesswork.
- **Action-Oriented Planning:** Unlike top-down approaches that can feel overwhelming, DFP naturally encourages an action-oriented mindset. Teams begin by defining the very first, tangible steps, which builds momentum and provides a clear starting point. The methodology continuously prompts the question, "What's the next action we need to take?" keeping the focus on forward progress rather than static planning.
- **Built-in Accountability and Reduced Errors:** The strict dependency rule is a powerful feature. By preventing team members from starting a parent task before its child tasks are complete, DFP proactively eliminates common project errors and rework. It creates a system of built-in accountability where progress on a task directly enables the next step for the team.
- **Improved Communication and Transparency:** The DFP canvas serves as a universal language for the project. Team members, stakeholders, and project managers can all look at the same visual map to understand the project's status, dependencies, and critical path at a glance. This high level of transparency fosters better communication, as everyone is working from the same, up-to-date plan. Other features of task nodes, such as tags on the task nodes to indicate the scope/domain of the task, the due indicator, node background colour to indicate which tasks are in-progress or overdue or upcoming, etc serves as a very helpful to get a crucial information about the tasks at a single glance.
- **Adaptability to Change:** While DFP is highly structured, it is also surprisingly adaptable. If a new task emerges or a dependency changes, the team can simply add a new node or a new line on the canvas. The rest of the map remains intact, and the new flow is immediately visible to everyone, allowing for quick adjustments without overhauling the entire plan. This makes DFP suitable for projects that may encounter scope changes.
- **Complementary and Flexible:** DFP does not impose a rigid, end-to-end framework on a project's lifecycle. Instead, it is designed to be a flexible, complementary tool that can be used at any phase. For instance, a team may use DFP solely for the initial planning and dependency mapping, then transition to a Kanban board for daily task management. Alternatively, the detailed schedule from a PERT chart can be implemented and maintained using DFP's living canvas view. This adaptability is most effective with digital tools that allow the same tasks to be viewed and managed through different methodological lenses, providing a synergistic benefit that traditional frameworks cannot.

3.2 Comparison with Existing Methodologies

Dependency Flow Planning (DFP) is not intended to replace existing methodologies but to complement them by providing a

foundational layer of sequential logic. While other frameworks excel in their specific functions, they often lack the integrated, bottom-up dependency management that DFP provides. A comparative analysis highlights DFP's unique position in the project management landscape.

This section's comparison with the Work Breakdown Structure (WBS) and Kanban is intended to highlight their limitations in managing certain project types, thereby justifying the need for DFP. It is not to suggest that DFP operates in isolation. In practice, DFP is designed to be a complementary methodology. The principles of WBS, for instance, can be utilized during the initial task creation phase of a project. Furthermore, DFP and Kanban can serve as synergistic companions, where DFP defines the strategic flow and dependencies, and Kanban manages the tactical, day-to-day execution. This dual-methodology approach can be effectively facilitated by a single digital tool [14].

A summary of the comparison of these methodologies is provided in Table I.

3.2.1 DFP vs. Work Breakdown Structure (WBS)

- **Planning Approach:** WBS is a top-down decomposition method, where a large project is broken into smaller parts. DFP is a bottom-up methodology, where a project is built by defining the next sequential step.
- **Focus:** A WBS focuses on scope and deliverables—what needs to be done. DFP focuses on flow and sequence—in what order things must be done.
- **Visual Representation:** The WBS is typically represented as a tree structure or an outline, which does not inherently show the order of work. The DFP canvas explicitly shows the flow with directional arrows and nodes.

3.2.2 DFP vs. Kanban

- **Primary Function:** Kanban is a tactical workflow management system focused on visualizing a process and limiting work in progress. DFP is a strategic planning methodology focused on mapping a project's sequential dependencies.
- **Dependency Management:** In Kanban, dependencies are often managed manually through notes or separate tools, and they are not a core part of the board's structure. DFP makes dependencies central and visible, with a strict child-to-parent rule that prevents tasks from starting prematurely.
- **Best Use Case:** Kanban is ideal for managing continuous, repetitive workflows (e.g., IT support tickets). DFP is best for projects with a clear tasks execution flow is required, a defined sequence of tasks (e.g., launching a product).

3.2.3 DFP vs. PERT Charts

- **Primary Focus:** PERT charts are primarily focused on managing uncertainty in project timelines. They use probabilistic estimations to determine a project's completion date. DFP is focused on the logical sequence of tasks and enforcing dependencies, regardless of a task's duration or associated uncertainty.
- **Planning Approach:** PERT charts require a top-down approach where all tasks and their dependencies must be defined upfront to perform the necessary statistical analysis. DFP uses a bottom-up, organic approach that allows the project map to be built incrementally as tasks are completed and new dependencies are discovered.
- **Ease of Use:** PERT charts can be complex to create and interpret due to the statistical calculations involved, often requiring specialized knowledge. The DFP canvas is designed to be intuitive and visually straightforward, making it accessible to all team members without formal training.

3.2.4 DFP vs. Gantt Charts

- **Primary Focus:** Gantt charts are schedule-centric. Their main purpose is to show when tasks will start and end, and how long they will take. DFP is logic-centric, focusing on the order of tasks regardless of their duration or specific dates.
- **Flexibility:** Gantt charts can be rigid and difficult to adjust, especially when project timelines change. DFP's canvas is highly flexible; tasks can be rearranged or new nodes added without disrupting the overall logical flow.
- **Ease of Use:** DFP's visual, map-based approach is often more intuitive and less intimidating for team members than a complex Gantt chart. DFP provides a clear understanding of the dependencies without requiring project managers to use specialized software.

3.3 Potential Use Cases

Dependency Flow Planning (DFP) is particularly well-suited for projects where the sequence of tasks is critical and where a clear, logical flow must be established and maintained. Its unique bottom-up approach and strict dependency enforcement make it ideal for a variety of industries and project types.

3.3.1 Software and Product Development

In software and product development, DFP can be used to plan the creation of a new feature or application. The design of a user interface must precede the front-end development, which in turn must precede integration testing. DFP can map this sequential flow, ensuring that developers don't begin work until the necessary design files are complete and approved. It can also be used to manage complex API development, where a new endpoint (parent task) cannot be built until all of its dependent services (child tasks) are functional.

3.3.2 Research and Development (R&D)

R&D projects are often defined by a series of experiments and discoveries, where the outcome of one stage dictates the next. DFP is a natural fit for this environment. A research team can use the canvas to map out their hypothesis, the experiment to test it, and the next steps based on the potential outcomes.

TABLE I
COMPARISON OF PROJECT MANAGEMENT METHODOLOGIES

Feature	Dependency Flow Planning (DFP)	Kanban	PERT Charts	Gantt Charts
Primary Goal	Define sequential flow & dependencies	Visualize & optimize workflow	Manage uncertain timelines	Plan & schedule timelines
Planning Approach	Bottom-Up	Continuous Flow	Top-Down	Top-Down
Dependency Handling	Central & Enforced	Manual/Tags	Built-in & Probabilistic	Built-in & Time-based
Visuals	Canvas Map (Nodes & Arrows)	Columns & Cards	Network Diagram	Bar Chart
Best For	Sequential, logical projects	Continuous, repetitive work	R&D & uncertain projects	Fixed-timeline projects

Each experiment is a child task that, when completed, unlocks the next phase of research. This allows for a flexible yet structured approach to scientific inquiry.

3.3.3 Event Planning and Management

Organizing a large event, such as a conference or a wedding, involves a complex web of dependencies. The venue cannot be booked until the budget is approved, and the catering cannot be finalized until the guest count is confirmed. DFP can visually map this entire process, ensuring that no critical task is missed and that all prerequisites are met before moving on to the next stage. It provides a clear, collaborative roadmap for all parties involved.

3.3.4 Manufacturing and Production

For projects in manufacturing and production, DFP can be used to plan the assembly process. The production of a component (child task) must be completed before it can be added to the final product (parent task). DFP provides a visual way to manage the supply chain and assembly line, ensuring that parts are ready and available at each stage of the process, reducing bottlenecks and improving efficiency.

3.4 Limitations and Challenges

Every methodology has its limitations, and Dependency Flow Planning (DFP) is no exception. While powerful for sequential projects, it may not be a perfect fit for every scenario. Understanding its challenges is crucial for effective implementation.

3.4.1 DFP's Limitations

- **Not Ideal for Non-Linear Projects:** DFP's strength—its strict sequential flow—is also its biggest limitation. For projects with highly iterative, non-linear, or parallel tasks that do not depend on one another, a DFP map can become overly complex and difficult to manage. For example, a creative brainstorming session for a marketing campaign might be better managed with a free-flowing mind map or a Kanban board, as there is no single "parent" deliverable.
- **Potential for Over-Planning:** The focus on defining every dependency can lead to analysis paralysis, especially at the project's outset. Teams might spend too much time mapping out every possible scenario instead of starting work. This is a risk in any planning-heavy methodology and must be mitigated with a clear scope and a focus on defining only the immediate, actionable next steps. For large-scale projects, it may be beneficial to use a methodology like CPM initially to identify the project's critical path. DFP can then be applied to manage the tasks along that chosen linear path, providing a focused approach to execution rather than comprehensive planning of every possible path.
- **Scalability for Very Large Projects:** While DFP can be used for large projects, a single, massive DFP canvas could become unwieldy. Managing thousands of interconnected nodes and lines might require advanced digital tools with features like sub-maps, collapsible nodes or view filters to maintain clarity. Without these features, the map could become a confusing "spaghetti chart."

3.4.2 Implementation Challenges

- **Tooling:** While the principles of DFP can be implemented with a simple whiteboard and sticky notes, a dedicated digital tool is necessary for remote teams and for managing large-scale projects. The market currently lacks a standardized, widely adopted software designed specifically for DFP's core principles. Most existing project management software would require custom setup or creative use of features to mimic the DFP workflow. At the time of writing this paper the only available option is Task Board plugin which is able to share the same tasks seamlessly across the DFP tool as well as Kanban board.
- **Cultural Shift:** Implementing DFP requires a shift in mindset for teams accustomed to top-down or non-linear planning. Team members must embrace the discipline of the "child-to-parent" flow and strictly adhere to the dependency rules. This requires training and a commitment from leadership to enforce the methodology.

Despite these challenges, DFP's value lies in its ability to bring a high level of clarity and discipline to projects where a clear, logical sequence is paramount. It is a powerful tool when applied to the right type of project with a team committed to its principles.

4 Conclusion

4.1 Summary of Key Findings

This paper has introduced and explored Dependency Flow Planning (DFP), a novel project management methodology designed to bring clarity and discipline to sequential projects. The key finding is that by reversing the traditional task hierarchy and focusing on a child-to-parent flow, DFP provides a unique and powerful alternative to existing frameworks. It fills a critical gap by making dependencies the central organizing principle of a project plan, a feature that is often an afterthought in other methodologies.

Our analysis demonstrated that DFP's core principles—the bottom-up flow, strict dependency enforcement, and a visual canvas—work together to create a robust and intuitive system. This approach leads to a more action-oriented and error-resistant planning process. Unlike the top-down nature of WBS or the time-centric focus of Gantt charts, DFP's logic-centric design provides a clear, living roadmap that is both easy to understand and highly adaptable. The methodology is best viewed as a complementary tool, used to strategically plan a project's sequence before its execution is managed with a tactical framework like Kanban.

4.2 Contributions to the Field

Dependency Flow Planning (DFP) makes several key contributions to the field of project management, offering a fresh perspective and a practical tool for practitioners.

First, it introduces a novel bottom-up planning paradigm. While top-down decomposition is the norm, DFP proves that projects can be effectively planned by building a roadmap from the ground up, starting with actionable tasks and linking them sequentially. This approach challenges conventional thinking and provides an alternative for projects that naturally evolve from a series of dependencies rather than a single, predefined end goal.

Second, DFP provides a unified framework for visual planning and dependency management. Many tools separate these two functions, forcing project managers to jump between a static hierarchy and a dynamic dependency chart. By integrating both on a single, intuitive canvas, DFP streamlines the planning process and ensures that logical flow is always visible and central to the project's state.

Finally, the methodology offers a flexible, yet disciplined approach. It combines the adaptability of visual mapping with the rigor of strict dependency rules. This hybrid model allows teams to manage complex projects with a high degree of confidence, knowing that the system will prevent common errors while remaining responsive to change. In essence, DFP provides a bridge between strategic planning and tactical execution, offering a new lens through which to view and manage work.

4.3 Future Work

The development of Dependency Flow Planning (DFP) opens several avenues for future research and practical application.

First, **empirical validation** [15] is needed to confirm the theoretical benefits of DFP. Case studies and comparative analyses with teams using other methodologies would provide quantitative data on DFP's impact on project timelines, efficiency, and team communication. This would also help identify the types of projects for which DFP is most effective.

Second, there is a significant opportunity for tooling development. The full potential of DFP can only be realized with a dedicated software platform. This tool would need features such as a dynamic canvas for creating and manipulating nodes and arrows, automated dependency enforcement, and integration with other project management software (like Kanban boards) to facilitate a seamless workflow.

Finally, further research can explore DFP's application in different contexts. How does DFP function in highly agile environments? Can its principles be adapted for personal productivity? Investigating these questions will expand the scope and utility of DFP, solidifying its place as a valuable addition to the project management toolkit.

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